Oceanic heat transport along the Norwegian Atlantic Current and the role of eddies

Key Points

- Eddies reduce 45% of heat from poleward oceanic heat toward Fram Strait
- Major eddy-induced heat leakage occurs near Lofoten Island and subsurface
- Eddy-induced heat leakage is relatively stable interannualy

Introduction

Background

- Warm Atlantic Water transports oceanic heat toward Arctic via Norwegian Atlantic Current
- Active mesoscale eddies observed in the Nordic Sea
- Eddies have been hypothesized to impact heat distribution
- Long-term Observation is scarce
- Previous numerical models are not sufficient to resolve mesoscale processes

Questions

- How much heat is carried by the current and how much reaches to the north?
- Would mesoscale eddies redistrbute heat? and how?

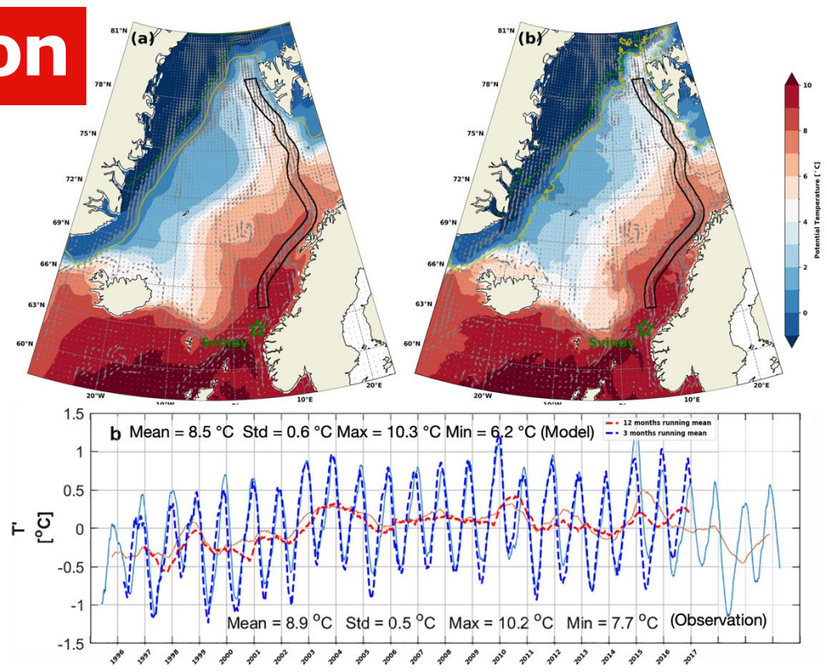

Fig. 1 (a) Observed and (b) modeled time-mean SST and surface geostrophic velocity for a representative year. Observed surface geostrophic velocity is derived from SSALTO/DUACS altimetry, while SST and sea ice data are obtained from OSTIA dataset. Green and yellow lines represent 30% monthly mean sea ice concentration isolines in March and September, respectively. The "worm-shaped" black line indicates study domain, and location of Svinøy station is marked. (c) Modeled temperature anomalies at 100 m depth nearSvinøy station, overlaid with corresponding observed measurements obtained from Orvik (2022). The observations are based on a 25-year average, while the modeled anomalies use a 21-year average, with 3- (blue line) and 12- (red line) months running mean applied.

Results

Table 1.1 **Heat budget (Terawatt) and decomposition (total=mean+eddy)**

Subdomain		Air-sea heat loss	Lateral advective heat				
			South	North	West	East	Convergence
Norwegian Shelf	Total	-6	112	-92	-32	18	6
	Mean		110	-92	-2	19	35
	Eddy		2	0	-30 (27%)	-1	-29
Norway to Svalbard	Total	-8	92	-53	46	-77	8
	Mean		92	-52	54	-72	22
	Eddy		0	-1	-8 (7%)	-5	-14
Svalbard Shelf	Total	-4	53	-35	-10	-4	4
	Mean		52	-35	-4	-3	10
	Eddy		1	0	-6 (5%)	-1	-6
Whole NwASC Domain	Total	-18	112	-35	4	-63	18
	Mean		110	-35	48	-56	67
	Eddy		2	0	-44 (39%)	-7	-49

1. Mean budget

- Eddy-induced westward leakage (44 TW) = 39% of the total heat supply (112 TW) = 2.5 times atmospheric losses (18 TW) > 1.3 time the heat reaching Fram Strait (35 TW)
- Westward eddy-induced heat transport counterbalances eastward mean advection

3. Depth & seasonal dependence

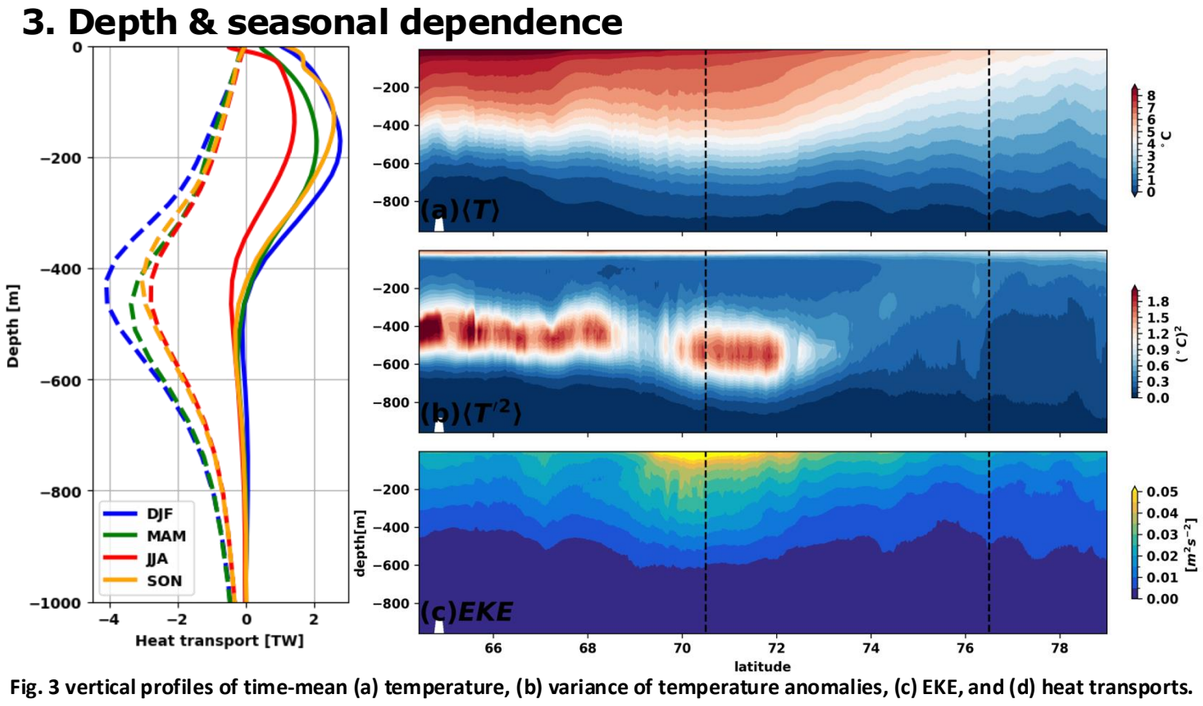

Fig. 3 vertical profiles of time-mean (a) temperature, (b) variance of temperature anomalies, (c) EKE, and (d) heat transports.

- Eddy heat transport peaks at intermediate depths (300–700 m), aligning with strong temperature variation near thermocline depth
- Heat transport exhibits wintertime intensification and summertime weakening

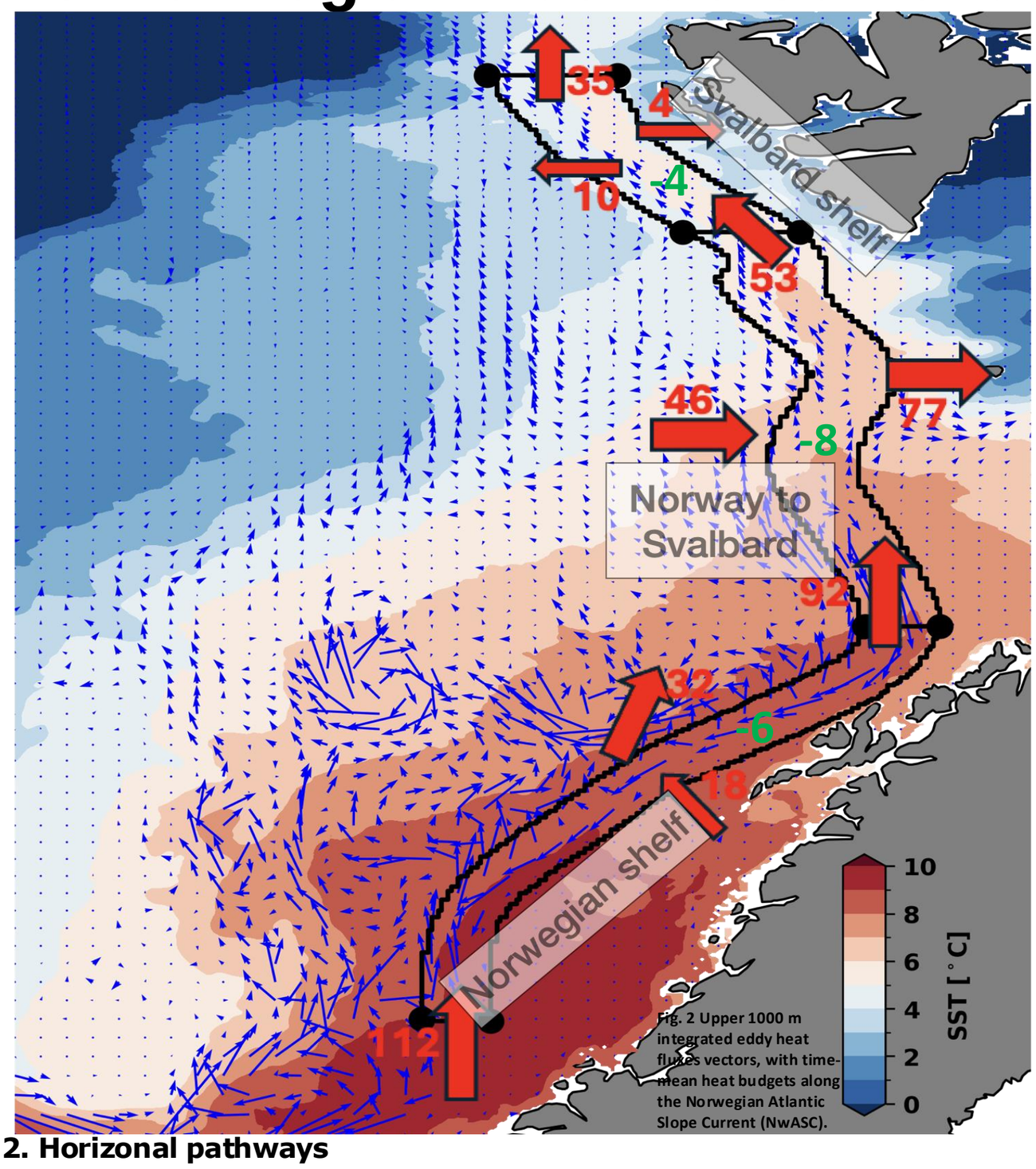

Fig. 2 Upper 1000 m integrated eddy heat fluxes vectors, with time-mean heat budgets along the Norwegian Atlantic Slope Current (NwASC).

2. Horizonal pathways

- Upper 1000 m integrated eddy heat fluxes follow two-branches pathways of NwAC
- Significant eddy heat fluxes extend laterally into inner basins
- A previously unrecognized counter-flow eddy heat pathway in the southern NwASC

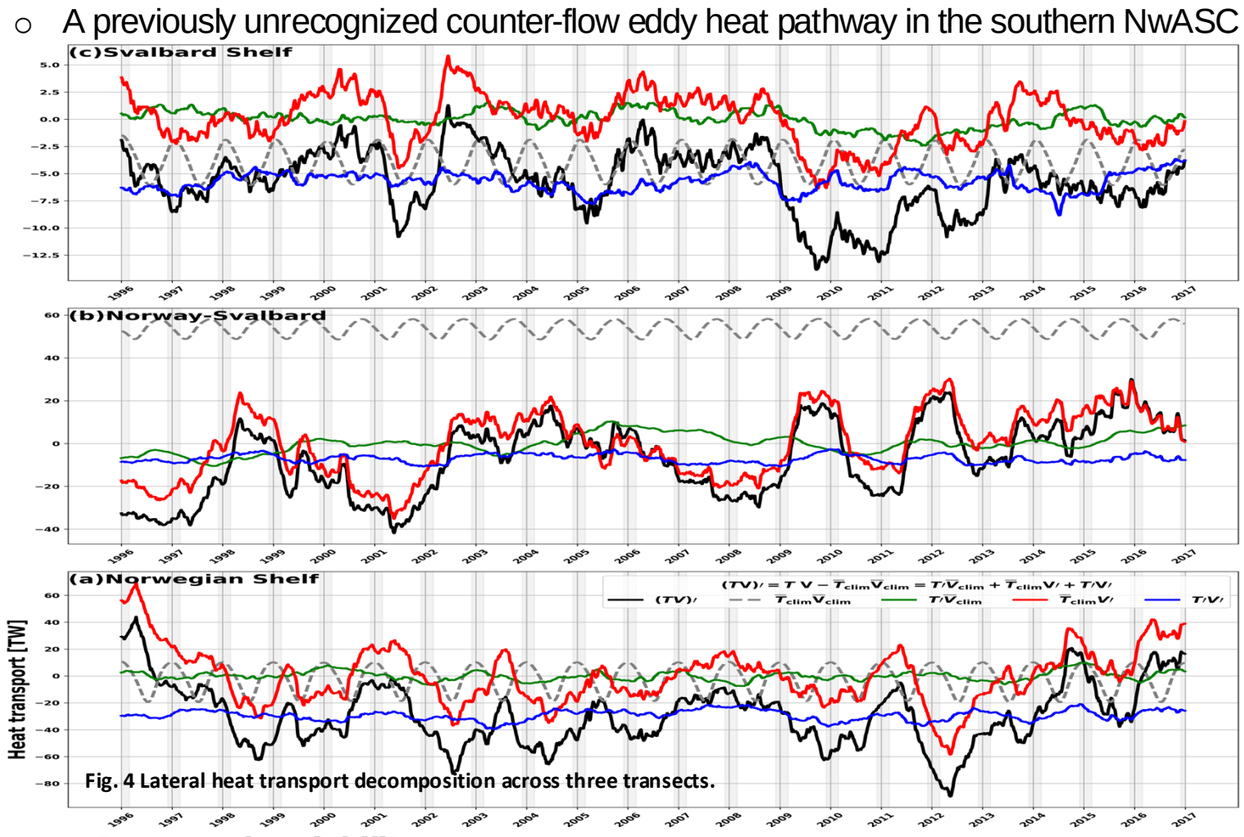

Fig. 4 Lateral heat transport decomposition across three transects.

4. Interannual variability

- Volume-driven heat transport exhibits strong interannual variability
- Eddy-driven transport remains consistently negative and relatively stable

Discussion

1. Eddy-induced heat redistribution

- Eddies divert 40% of total heat from the southern supply agrees with observational estimate (Bashmachnikov et al., 2023).
- Eddy heat fluxes from the Lofoten Escarpment sustains Lofoten Vortex (Isachsen et al., 2012).
- a counter-flow eddy heat pathway in the southern NwASC diminishes northward heat.

2. Volume- and eddy driven heat transport

- Advective heat anomalies are dominated by velocity variations, in agreement with upstream variability (Årthun and Eldevik, 2016).
- Relatively stable eddy heat leakage suggests that fluctuations in upstream transport are insufficient to explain mesoscale variability on interannual time scale.

Conclusions

- Approximately 112 TW heat is carried by NwASC, with 35 TW reaching Fram Strait.
- Eddy-induced lateral heat leakage along NwASC (-49TW) significantly reduces poleward heat transport to the Arctic, peaking near Lofoten Escarpment at mid-depths.
- Cooling of the current arises not only from lateral leakage but also from anti-stream eddy heat fluxes associated along-stream thermocline gradients, which may diminish northward propagating heat anomalies.
- Although winter intensifies leakage, its interannual stability highlights that mesoscale processes are not only subject to the current strength.
- Local atmospheric heat loss is minor; instead, lateral eddy fluxes redistribute massive heat into the broad Norwegian Sea.

Methods

- High-resolution model (MITgcm) validated by satellite and in-situ measurements
- Heat budget analysis: $\underbrace{\frac{\partial H}{\partial t}}_{\text{Tend}} = \underbrace{\rho C_p \int (-\nabla \cdot T\mathbf{V})\, dA}_{\text{Adv}} + \underbrace{\int q\, dS}_{\text{Qnet}}$
- Reynolds Decomposition: $T = \overline{T}_{clim} + T' \quad \mathbf{V} = \overline{\mathbf{V}}_{clim} + \mathbf{V}'$

$$Q_{\text{total}} = \overline{T}_{\text{clim}}\,\overline{\mathbf{V}}_{\text{clim}} + \overline{T}_{\text{clim}}\mathbf{V}' + T'\overline{\mathbf{V}}_{\text{clim}} + T'\mathbf{V}'$$

References

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